Ryan Dolan Rodriguez

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EDUCATION

University of Notre Dame | Notre Dame, Indiana Expected Graduation Date: May 2028
Bachelor of Science, Mechanical Engineering | Minor: Engineering Corporate Practice GPA: 3.8/4.0
Engineering Study Abroad | Alcoy, Spain | Universidad Politécnica de Valencia Summer 2025

ENGINEERING PROJECT EXPERIENCE

CubeSat | Notre Dame, IN August 2025 – Present
IRIS Design Team Member

- Supported reliable payload deployment above 89,000 ft by designing and validating 3D CAD components, flight systems, and deployment mechanisms for a long-duration near-space payload
- Reduced per-unit hardware cost to approximately $1,200 by driving mechanical development from prototyping through fabrication, personally manufacturing structural components using CNC machining, waterjet cutting, and 3D printing and coordinating machined aluminum chassis production to specification
- Validated mission readiness above 89,000 ft by designing and executing a hardware measurement and test plan across thermal, structural, and electrical subsystems and presenting results in a formal engineering design review
- Reduced aerodynamic drag uncertainty by 30%+ on flight-critical deployment mechanisms by performing Ansys CFD simulations to characterize airflow profiles and validate structural load assumptions
- Eliminated structural failure risk on a 5.5 lb near-space payload by performing Ansys FEA to evaluate stress, deformation, and factor of safety under aerodynamic and deployment loads
- Enabled 2.5+ hours of thermal survivability at $-10°C$ by integrating servo controllers, GPS sensors, and attitude sensors into a custom PCB and coordinating power across heating, avionics, and actuation systems

Satellite Payload Vibration Isolation | Notre Dame, IN August 2025 – December 2025
Independent Project

- Validated a spring-mass-damper model of the SpaceX Falcon Heavy launch environment by matching MATLAB numerical simulations to closed-form analytical solutions across multiple base-excitation test cases.
- Minimized peak transmitted acceleration on a 2,982 kg satellite payload by designing an optimal vibration isolation system, benchmarking Moog SoftRide UniFlex isolator configurations against measured performance across a 5–100 Hz sine-sweep test environment.

INTERNSHIP EXPERIENCE

Industry Labs | South Bend, IN May 2026 – August 2026
Technical Consulting Intern

- Improved engineering specification retrieval to 83.5% accuracy with zero hallucinations across a 103-question validation set by developing and testing a technical knowledge system for a $1.3B+ OEM supplier's product engineering team
- Recovered an estimated 3,000 engineering hours and $200K+ annually across 60–70 product and design engineers by benchmarking system performance against existing mechanical specification and design-revision workflows
- Restored access to 42% of a previously unsearchable engineering document library by developing a workflow that processed scanned product specifications and engineering drawings 6× faster than the manual baseline

LEADERSHIP EXPERIENCE

Society of Hispanic Professional Engineers (SHPE) | Notre Dame, IN August 2024 – Present
Conference Chair

- Led planning and execution for a 40+ person national conference delegation by managing travel, logistics, fundraising, and a multi-thousand-dollar chapter budget
- Raised $10,000+ in conference funding by organizing football gameday fundraisers, enabling travel and registration support for the chapter delegation

TECHNICAL AND LANGUAGE SKILLS

CAD/CAM: SolidWorks, Autodesk Fusion 360, PTC CREO, Geometric Dimensioning & Tolerancing (GD&T)
Simulation: Ansys (FEA, CFD), MATLAB
Fabrication: CNC machining, Waterjet Cutting, 3D Printing, Injection Molding
Programming: Python, MATLAB, Microsoft Excel, Microsoft Word, Microsoft Office Products
Languages: English (fluent), Spanish (fluent)